

EE3621: Digital Signal Processing

Lecture 17: IIR Filter — Direct Form I and Direct Form II Realization (III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 06:00 (6 mins):** Introduction to IIR Realization: Difference equations, rational transfer functions $H(z) = B(z)/A(z)$, and recursive feedback loops.
 - **06:00 – 16:00 (10 mins):** Direct Form I Realization: Derivation, block diagram, feed-forward zeros followed by feedback poles, and non-canonical delay count $(M + N)$.
 - **16:00 – 26:00 (10 mins):** Direct Form II (Canonical Realization): The LTI Commutativity Principle (swapping zeros and poles), intermediate state variable $w[n]$, and merging redundant delay registers.
 - **26:00 – 33:00 (7 mins):** Side-by-side comparison: Direct Form I vs. Direct Form II (Hardware complexity, memory footprint, and internal node overflow risks).
 - **33:00 – 38:00 (5 mins):** Step-by-step worked numerical examples (2nd-order biquad and asymmetric order systems).
 - **38:00 – 40:00 (2 mins):** Pedagogical checkpoints, common sign misconceptions, and university exam review.
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2 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Formulate** the general linear constant-coefficient difference equation (LCCDE) and its corresponding rational transfer function $H(z)$ for an arbitrary IIR filter.
2. **Derive** the mathematical state equations and signal flow graphs (SFG) for the Direct Form I realization.
3. **Apply** the LTI Commutativity Principle to reverse the order of the all-pole and all-zero sub-systems without altering the overall system response.
4. **Formulate** the intermediate state variable $w[n]$ to construct the canonical Direct Form II realization.
5. **Calculate** exact hardware resource counts (multipliers, adders, and delay registers) for both canonical $(\max(M, N))$ and non-canonical $(M + N)$ structures.

6. **Compare** Direct Form I and Direct Form II in terms of memory requirements, critical path latency, and intermediate register overflow sensitivity.
 7. **Extract** difference equations and transfer functions directly from block diagrams and signal flow graphs.
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3 2. MATHEMATICAL FOUNDATIONS OF IIR SYSTEMS

3.1 2.1 Difference Equations and Rational Transfer Functions

An Infinite Impulse Response (IIR) digital filter is described by a linear constant-coefficient difference equation (LCCDE) relating the input sequence $x[n]$ to the output sequence $y[n]$:

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k] \quad (1)$$

Without loss of generality, we normalize the current output coefficient by setting $a_0 = 1$. Rearranging into computational form:

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k] \quad (2)$$

Expanding this expression explicitly:

$$y[n] = \underbrace{b_0 x[n] + b_1 x[n-1] + \cdots + b_M x[n-M]}_{\text{Feedforward Terms (Zeros)}} - \underbrace{(a_1 y[n-1] + a_2 y[n-2] + \cdots + a_N y[n-N])}_{\text{Feedback Terms (Poles)}} \quad (3)$$

Applying the unilateral or bilateral Z -transform to both sides (assuming zero initial conditions):

$$Y(z) = \sum_{k=0}^M b_k z^{-k} X(z) - \sum_{k=1}^N a_k z^{-k} Y(z) \quad (4)$$

Gathering $Y(z)$ terms on the left:

$$Y(z) \left(1 + \sum_{k=1}^N a_k z^{-k} \right) = X(z) \left(\sum_{k=0}^M b_k z^{-k} \right) \quad (5)$$

The system transfer function $H(z)$ is the ratio of output $Y(z)$ to input $X(z)$:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} = \frac{B(z)}{A(z)} \quad (6)$$

where:

- $B(z) = \sum_{k=0}^M b_k z^{-k}$ is the numerator polynomial of degree M , determining the **system zeros**.
- $A(z) = 1 + \sum_{k=1}^N a_k z^{-k}$ is the denominator polynomial of degree N , determining the **system poles**.
- The order of the filter is defined as $\max(M, N)$, typically with $M \leq N$.

3.2 2.2 Critical Pedagogical Note on Feedback Sign Conventions

Faculty Insight & Exam Caution: Pay meticulous attention to the minus sign in front of the feedback coefficients a_k !

- In the difference equation (2), the feedback terms are **subtracted**: $-\sum_{k=1}^N a_k y[n-k]$.
- In the denominator polynomial $A(z) = 1 + \sum_{k=1}^N a_k z^{-k}$, the coefficients a_k have **positive signs**.
- When drawing block diagrams and signal flow graphs:
 - If the summing junction performs **addition** (+), the multiplier branch must be labeled $-\mathbf{a_k}$.
 - If the multiplier branch is labeled $+\mathbf{a_k}$, the summing junction must have a **subtraction** (−) sign at that input.

Confusing this sign is the single most common student error in university semester examinations.

4 3. DIRECT FORM I REALIZATION (NON-CANONICAL)

4.1 3.1 Derivation from Sub-System Factoring

The transfer function $H(z)$ in (6) can be factored into a cascade of two independent LTI sub-systems:

$$H(z) = H_1(z) \cdot H_2(z) = B(z) \cdot \frac{1}{A(z)} \quad (7)$$

Let $v[n]$ (with Z -transform $V(z)$) represent the intermediate signal between the two stages:

$$X(z) \longrightarrow \boxed{H_1(z) = B(z)} \xrightarrow{V(z)} \boxed{H_2(z) = \frac{1}{A(z)}} \longrightarrow Y(z) \quad (8)$$

4.1.1 Stage 1: All-Zero (Feedforward / FIR) Sub-System

$$V(z) = B(z)X(z) = \left(\sum_{k=0}^M b_k z^{-k} \right) X(z) \quad (9)$$

Taking the inverse Z -transform yields the feedforward difference equation:

$$v[n] = \sum_{k=0}^M b_k x[n-k] = b_0 x[n] + b_1 x[n-1] + \dots + b_M x[n-M] \quad (10)$$

This stage is a pure FIR tapped-delay-line structure requiring M delay elements (z^{-1}) to store past inputs $x[n-1], \dots, x[n-M]$.

4.1.2 Stage 2: All-Pole (Feedback / Autoregressive) Sub-System

$$Y(z) = \frac{1}{A(z)} V(z) \implies A(z)Y(z) = V(z) \quad (11)$$

$$\left(1 + \sum_{k=1}^N a_k z^{-k} \right) Y(z) = V(z) \implies Y(z) = V(z) - \sum_{k=1}^N a_k z^{-k} Y(z) \quad (12)$$

Taking the inverse Z -transform:

$$y[n] = v[n] - \sum_{k=1}^N a_k y[n-k] = v[n] - a_1 y[n-1] - a_2 y[n-2] - \dots - a_N y[n-N] \quad (13)$$

This stage is a pure recursive all-pole structure requiring N delay elements (z^{-1}) to store past outputs $y[n-1], \dots, y[n-N]$.

4.2 3.2 Direct Form I Hardware Complexity Audit

By cascading Stage 1 followed by Stage 2, we obtain the ****Direct Form I Realization****:

- **Delay Elements** (z^{-1}): M delays for input $x[n]$ plus N delays for output $y[n]$:

$$\text{Total Delays} = M + N \quad (14)$$

- **Multipliers**: $M + 1$ feedforward multipliers ($b_0, b_1, \dots, b_M$) plus N feedback multipliers ($a_1, a_2, \dots, a_N$):

$$\text{Total Multipliers} = M + N + 1 \quad (15)$$

- **Adders**: M additions for feedforward accumulation plus N additions for feedback accumulation:

$$\text{Total Adders} = M + N \quad (16)$$

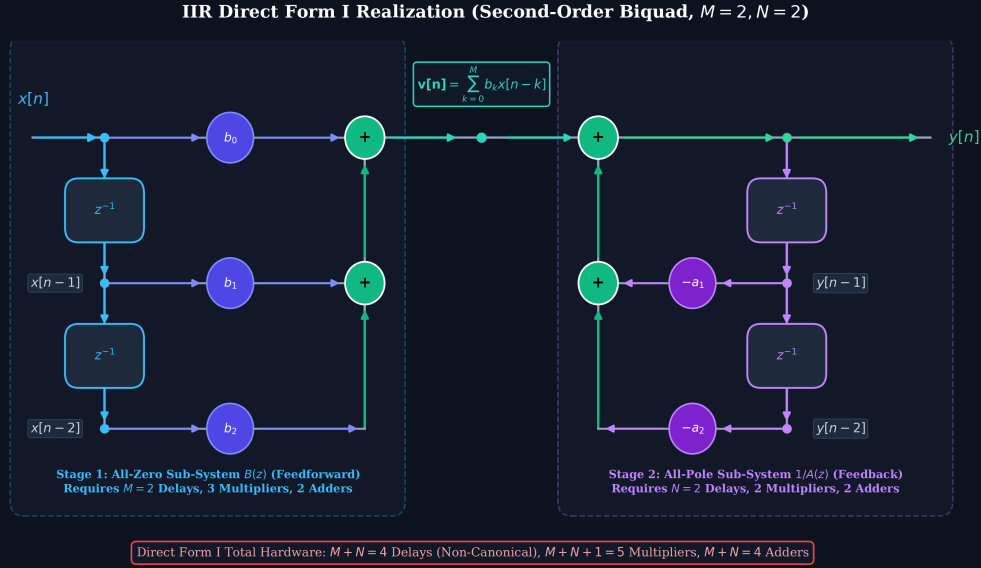


Figure 1: IIR Direct Form I Realization Schematic. The input delay line (left, M delays) computes the all-zero feedforward response, while the output delay line (right, N delays) computes the all-pole feedback response.

4.3 3.3 Why Direct Form I is Non-Canonical

A digital filter realization is formally defined as ****canonical**** if the number of delay registers (z^{-1}) used in the hardware realization is strictly equal to the order of the filter:

$$\text{Canonical Delay Requirement} = \max(M, N) \quad (17)$$

In Direct Form I, the total number of delay elements is $M + N$. Because $M + N > \max(M, N)$ for any non-trivial filter ($M, N \geq 1$), ****Direct Form I is strictly non-canonical****. For a typical system where $M = N$, Direct Form I consumes ****twice as many memory registers**** as mathematically required!

5 4. DIRECT FORM II REALIZATION (CANONICAL)

5.1 4.1 The LTI Commutativity Principle

Linear Time-Invariant (LTI) systems possess the fundamental mathematical property of ****commutativity under convolution****:

$$h_1[n] * h_2[n] = h_2[n] * h_1[n] \iff H(z) = H_1(z) \cdot H_2(z) = H_2(z) \cdot H_1(z) \quad (18)$$

In Direct Form I, we implemented $H(z) = B(z) \cdot \frac{1}{A(z)}$ (zeros first, then poles). Because the sub-systems commute, we can completely **reverse the order of cascade** without altering the overall input-output behavior:

$$H(z) = \frac{1}{A(z)} \cdot B(z) \quad (19)$$

$$X(z) \longrightarrow \boxed{H_2(z) = \frac{1}{A(z)}} \xrightarrow{W(z)} \boxed{H_1(z) = B(z)} \longrightarrow Y(z) \quad (20)$$

We now define an intermediate state variable $w[n]$ (with transform $W(z)$) at the junction between the all-pole sub-system and the all-zero sub-system.

5.2 4.2 Mathematical Derivation of State Equations

5.2.1 Stage 1: The All-Pole Sub-System

The input $X(z)$ is processed by the all-pole transfer function $1/A(z)$ to generate the intermediate state sequence $W(z)$:

$$W(z) = \frac{1}{A(z)} X(z) \implies A(z)W(z) = X(z) \quad (21)$$

Substituting $A(z) = 1 + \sum_{k=1}^N a_k z^{-k}$:

$$\left(1 + \sum_{k=1}^N a_k z^{-k}\right) W(z) = X(z) \implies W(z) = X(z) - \sum_{k=1}^N a_k z^{-k} W(z) \quad (22)$$

Taking the inverse Z-transform yields the ****State Update Equation****:

$$w[n] = x[n] - \sum_{k=1}^N a_k w[n-k] = x[n] - a_1 w[n-1] - a_2 w[n-2] - \dots - a_N w[n-N] \quad (23)$$

5.2.2 Stage 2: The All-Zero Sub-System

The intermediate sequence $W(z)$ is then processed by the all-zero transfer function $B(z)$ to produce the final output $Y(z)$:

$$Y(z) = B(z)W(z) = \left(\sum_{k=0}^M b_k z^{-k}\right) W(z) \quad (24)$$

Taking the inverse Z-transform yields the ****Output Equation****:

$$y[n] = \sum_{k=0}^M b_k w[n-k] = b_0 w[n] + b_1 w[n-1] + b_2 w[n-2] + \dots + b_M w[n-M] \quad (25)$$

5.3 4.3 The Delay Merging Breakthrough (Canonical Architecture)

Examine equations (23) and (25) simultaneously:

- Equation (23) requires the delayed samples: $w[n-1], w[n-2], \dots, w[n-N]$.
- Equation (25) requires the delayed samples: $w[n-1], w[n-2], \dots, w[n-M]$.

The Crucial Architectural Discovery: Both the feedback branch (poles) and the feedforward branch (zeros) operate on the **identical sequence of delayed values of $w[n]$** !

Therefore, we do not need two separate delay lines. We can implement a **single, shared central delay line** containing:

$$\text{Total Delays Required} = \max(M, N) \quad (26)$$

The delayed states $w[n-k]$ are tapped:

1. Toward the **left** with scaling factors $-a_k$ to feed the feedback summation generating $w[n]$.
2. Toward the **right** with scaling factors b_k to feed the feedforward summation generating $y[n]$.

Because the number of delay registers equals the system order $\max(M, N)$, ****Direct Form II is strictly canonical****.

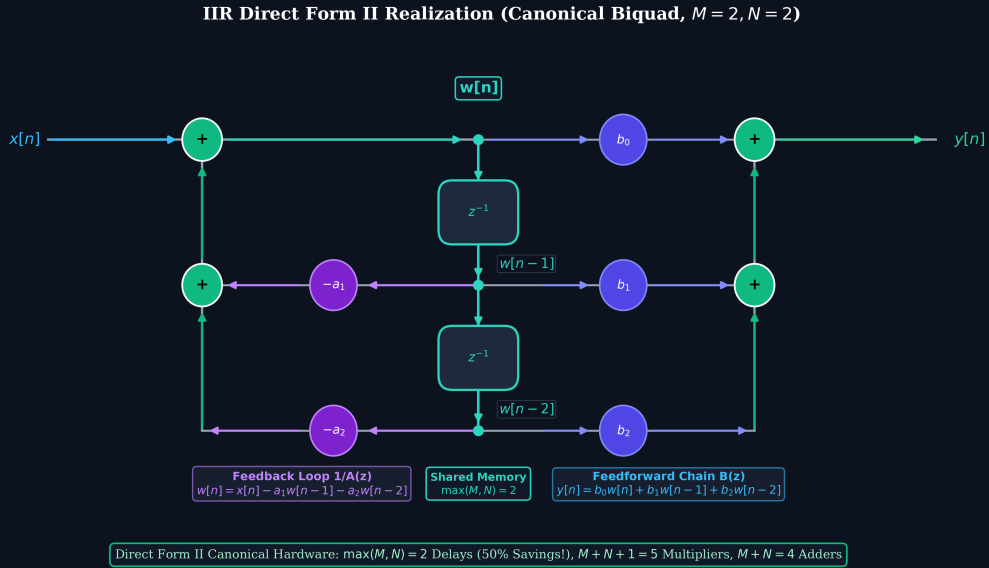


Figure 2: IIR Direct Form II (Canonical) Realization Schematic. A single central delay line of length $\max(M, N)$ is shared between the feedback all-pole loop (left) and the feedforward all-zero chain (right).

6 5. DIRECT FORM I VS. DIRECT FORM II COMPARISON

6.1 5.1 Comprehensive Hardware Resource Comparison Table

Architectural Metric	Direct Form I (DF-I)	Direct Form II (DF-II)
Canonical Status	Non-Canonical	Canonical
Total Delay Elements (z^{-1})	$M + N$	$\max(M, N)$ (50% Savings when $M = N$)
Total Multipliers	$M + N + 1$	$M + N + 1$ (Identical)
Total Adders	$M + N$	$M + N$ (Identical)
Delay Allocation	Separated (M for x , N for y)	Shared central delay line (w)
Signal Access	Direct access to input/output	Requires internal state $w[n]$
Dynamic Range Overflow	Low internal node gain	High internal node gain at $w[n]$
Critical Path Latency	1 Multiply + $\max(M, N)$ Adds	1 Multiply + $(M + N)$ Adds

Table 1: Comprehensive Comparison between Direct Form I and Direct Form II Realizations.

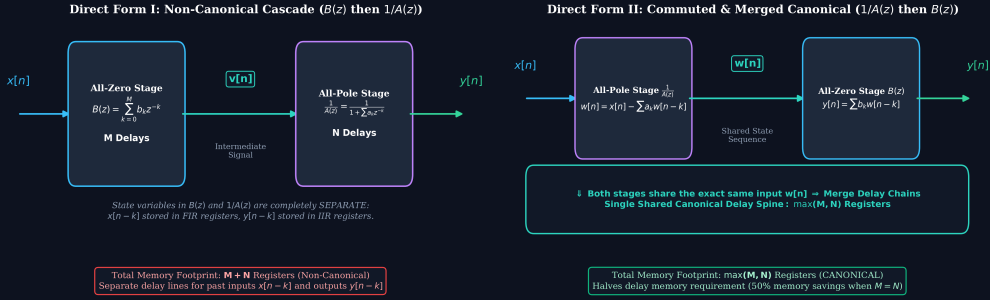


Figure 3: Architectural Summary: Direct Form I (separate delay lines, high memory footprint) vs. Direct Form II (shared central delay line, minimal memory footprint).

6.2 5.2 The Critical Engineering Trade-off: Overflow Sensitivity

While Direct Form II is universally preferred in textbooks for its memory efficiency, practical DSP hardware engineers must understand its primary limitation: **internal node dynamic range overflow**.

- In Direct Form II, the transfer function from input $x[n]$ to intermediate state $w[n]$ is:

$$H_{xw}(z) = \frac{W(z)}{X(z)} = \frac{1}{A(z)} = \frac{1}{1 + \sum_{k=1}^N a_k z^{-k}} \quad (27)$$

- If the filter has high- Q poles (poles close to the unit circle, $|p_k| \approx 1$, as in narrowband low-pass or high-selectivity band-pass filters), the gain $|H_{xw}(e^{j\omega})|$ can be **extremely large** (e.g., 40 dB to 60 dB, or a voltage amplification of 100 to 1000).
- Even if the overall filter gain $|H(e^{j\omega})| \leq 1$, the internal node $w[n]$ will violently overflow standard fixed-point 16-bit or 24-bit registers, causing catastrophic limit-cycle oscillations and gross audio distortion.
- In contrast, Direct Form I realizes the zeros $B(z)$ first, which naturally attenuates input signals before they reach the high-gain feedback loop.

7 6. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

7.1 Example 17.1: Second-Order Biquad Realization (Complete Derivation)

Problem: A second-order IIR digital filter is described by the transfer function:

$$H(z) = \frac{2 + 3z^{-1} + 4z^{-2}}{1 - 0.6z^{-1} + 0.25z^{-2}} \quad (28)$$

- (a) Identify all filter coefficients and state the filter order.
- (b) Formulate the Direct Form I difference equations and calculate hardware resource counts.
- (c) Derive the Direct Form II state update and output difference equations.
- (d) Construct the complete signal flow graph for both forms.
- (e) Compute the impulse response $y[n]$ for $n = 0, 1, 2$ using Direct Form II equations.

Solution:

(a) **Coefficient Identification & Order:** Comparing with the general rational form:

$$H(z) = \frac{b_0 + b_1z^{-1} + b_2z^{-2}}{1 + a_1z^{-1} + a_2z^{-2}} \quad (29)$$

We extract:

- Numerator coefficients: $b_0 = 2, b_1 = 3, b_2 = 4$ ($M = 2$)
- Denominator coefficients: $a_1 = -0.6, a_2 = +0.25$ ($N = 2$)
- Filter Order: $\max(M, N) = \max(2, 2) = \mathbf{2}$ (Second-Order Biquad).

(b) **Direct Form I Realization:** The overall difference equation is:

$$y[n] = 2x[n] + 3x[n-1] + 4x[n-2] - \left(-0.6y[n-1] + 0.25y[n-2] \right) \quad (30)$$

$$y[n] = 2x[n] + 3x[n-1] + 4x[n-2] + 0.6y[n-1] - 0.25y[n-2] \quad (31)$$

Hardware Resource Audit for DF-I:

- Delay elements (z^{-1}): $M + N = 2 + 2 = \mathbf{4}$ delays ($x[n-1], x[n-2], y[n-1], y[n-2]$).
- Multipliers: $M + N + 1 = 2 + 2 + 1 = \mathbf{5}$ multipliers ($2, 3, 4, 0.6, -0.25$).
- Adders: $M + N = 2 + 2 = \mathbf{4}$ two-input adders.

(c) **Direct Form II Realization (Canonical):** We define the intermediate sequence $w[n]$:

$$\begin{aligned} w[n] &= x[n] - a_1w[n-1] - a_2w[n-2] \\ w[n] &= x[n] + 0.6w[n-1] - 0.25w[n-2] \quad \textbf{(State Equation)} \end{aligned} \quad (32)$$

The output equation evaluates the feedforward chain on $w[n]$:

$$y[n] = 2w[n] + 3w[n-1] + 4w[n-2] \quad \textbf{(Output Equation)} \quad (33)$$

Hardware Resource Audit for DF-II:

- Delay elements (z^{-1}): $\max(M, N) = \max(2, 2) = \mathbf{2}$ delays ($w[n-1], w[n-2]$).
- Multipliers: $M + N + 1 = \mathbf{5}$ multipliers.
- Adders: $M + N = \mathbf{4}$ adders (2 in the feedback sum, 2 in the output sum).

- **Savings:** Delay memory cut by **50%** (from 4 down to 2)!

(e) **Step-by-Step Impulse Response Evaluation** ($x[n] = \delta[n]$): Assume filter is initially relaxed ($w[-1] = 0, w[-2] = 0$):

- **At $n = 0$ ($x[0] = 1$):**

$$\begin{aligned} w[0] &= x[0] + 0.6w[-1] - 0.25w[-2] = 1 + 0 - 0 = \mathbf{1.0} \\ y[0] &= 2w[0] + 3w[-1] + 4w[-2] = 2(1.0) + 0 + 0 = \mathbf{2.0} \end{aligned}$$

- **At $n = 1$ ($x[1] = 0$):**

$$\begin{aligned} w[1] &= x[1] + 0.6w[0] - 0.25w[-1] = 0 + 0.6(1.0) - 0 = \mathbf{0.6} \\ y[1] &= 2w[1] + 3w[0] + 4w[-1] = 2(0.6) + 3(1.0) + 0 = 1.2 + 3.0 = \mathbf{4.2} \end{aligned}$$

- **At $n = 2$ ($x[2] = 0$):**

$$\begin{aligned} w[2] &= x[2] + 0.6w[1] - 0.25w[0] = 0 + 0.6(0.6) - 0.25(1.0) = 0.36 - 0.25 = \mathbf{0.11} \\ y[2] &= 2w[2] + 3w[1] + 4w[0] = 2(0.11) + 3(0.6) + 4(1.0) = 0.22 + 1.8 + 4.0 = \mathbf{6.02} \end{aligned}$$

The first three impulse response samples are $h[n] = \{\mathbf{2.0}, 4.2, 6.02, \dots\}$.

7.2 Example 17.2: Third-Order Asymmetric System ($M \neq N$)

Problem: Realize the third-order IIR system:

$$H(z) = \frac{1 + 0.5z^{-1}}{1 - 1.2z^{-1} + 0.7z^{-2} - 0.1z^{-3}} \quad (34)$$

in (a) Direct Form I and (b) Direct Form II.

Solution: Here the degrees are $M = 1$ (numerator) and $N = 3$ (denominator). Coefficients:

$$b_0 = 1, \quad b_1 = 0.5, \quad b_2 = 0, \quad b_3 = 0 \quad (35)$$

$$a_1 = -1.2, \quad a_2 = 0.7, \quad a_3 = -0.1 \quad (36)$$

(a) **Direct Form I:**

$$y[n] = x[n] + 0.5x[n-1] + 1.2y[n-1] - 0.7y[n-2] + 0.1y[n-3] \quad (37)$$

- Requires $M + N = 1 + 3 = 4$ delay registers (z^{-1}).
- Multipliers: 2 (zeros) + 3 (poles) = **5** multipliers.
- Adders: 1 (zeros) + 3 (poles) = **4** adders.

(b) **Direct Form II (Canonical):**

$$w[n] = x[n] + 1.2w[n-1] - 0.7w[n-2] + 0.1w[n-3] \quad (38)$$

$$y[n] = w[n] + 0.5w[n-1] \quad (39)$$

- Shared delay line length: $\max(M, N) = \max(1, 3) = \mathbf{3}$ delays ($w[n-1], w[n-2], w[n-3]$).
- Multipliers: **5** multipliers.
- Adders: **4** adders.
- Delay savings: Reduced from 4 to 3 registers (25% savings).

8 7. PEDAGOGICAL CHECKPOINTS & COMMON MIS- CONCEPTIONS

1. Misconception 1: "Does Direct Form II change the zeros, poles, or frequency response of the filter?"

- *Reality: No.* Under exact mathematical precision (infinite word-length), Direct Form I and Direct Form II have mathematically identical transfer functions $H(z)$ and generate identical output sequences $y[n]$. The transformation relies solely on the commutativity of linear time-invariant systems ($H_1H_2 = H_2H_1$).

2. Misconception 2: "Why are the feedback multiplier coefficients negative ($-a_k$)?"

- *Reality:* When an LTI difference equation is written in output-explicit form:

$$y[n] = \sum b_k x[n-k] - \sum a_k y[n-k]$$

the feedback terms appear with a negative sign. If the hardware summing junction is purely additive (+), each feedback coefficient must carry a sign inversion ($-a_k$).

3. Misconception 3: "Is Direct Form II always better than Direct Form I because it uses fewer delays?"

- *Reality: Not necessarily.* While Direct Form II saves silicon area by halving the register count, its internal state node $w[n] = x[n]/A(z)$ can exhibit severe amplitude peaking. In fixed-point hardware, Direct Form II frequently suffers from catastrophic arithmetic overflow, whereas Direct Form I attenuates input signals first.

4. Misconception 4: "Can high-order IIR filters ($N \geq 8$) be safely implemented in Direct Form II?"

- *Reality: Never in practice.* For orders $N > 4$, the roots of high-degree polynomials are hypersensitive to tiny coefficient quantization errors. A 1% error in an a_k coefficient can push poles outside the unit circle ($|z| > 1$), causing instability. In commercial DSP systems, high-order IIR filters are always factored into a cascade or parallel arrangement of second-order biquads (SOS).

9 8. UNIVERSITY EXAMINATION QUESTIONS & PRACTICE PROBLEMS

9.1 Problem 1 (10 Marks — Analytical & Realization)

Question: An IIR digital filter is governed by the difference equation:

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n] + \frac{1}{3}x[n-1] \quad (40)$$

1. Determine the system transfer function $H(z)$, its poles, and its zeros. (3 Marks)
2. Construct the Direct Form I signal flow graph and state the exact hardware count. (3 Marks)
3. Construct the Direct Form II canonical signal flow graph and evaluate the percentage reduction in delay registers. (4 Marks)

Marking Scheme & Solution Outline:

1. **Transfer Function & Roots:**

$$H(z) = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}} = \frac{1 + \frac{1}{3}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})} \quad (41)$$

Zero at $z = -1/3$. Poles at $p_1 = 1/2 = 0.5$ and $p_2 = 1/4 = 0.25$. Both poles lie inside the unit circle $\implies$ System is Stable. (3 Marks)

2. **Direct Form I:** $y[n] = x[n] + \frac{1}{3}x[n-1] + \frac{3}{4}y[n-1] - \frac{1}{8}y[n-2]$.

Delays: $M + N = 1 + 2 = \mathbf{3}$ delays. Multipliers: $2 + 2 = 4$ (or 3 non-unity). Adders: $\mathbf{3}$ adders. (3 Marks)

3. **Direct Form II (Canonical):**

$$\begin{aligned} w[n] &= x[n] + \frac{3}{4}w[n-1] - \frac{1}{8}w[n-2] \\ y[n] &= w[n] + \frac{1}{3}w[n-1] \end{aligned}$$

Delays: $\max(1, 2) = \mathbf{2}$ delays. Multipliers: 4. Adders: 3.

Delay savings: $\frac{3-2}{3} \times 100\% = \mathbf{33.33\%}$ reduction. (4 Marks)

9.2 Problem 2 (5 Marks — Conceptual Comparison)

Question: Explain what is meant by a "canonical structure" in digital signal processing. Prove why Direct Form II is canonical while Direct Form I is non-canonical. Under what conditions might an engineer deliberately choose Direct Form I over Direct Form II?

Solution Outline:

- **Definition:** A filter realization is canonical if the number of delay registers (z^{-1}) equals the order of the transfer function, defined as $\max(M, N)$. (1 Mark)
- **Proof:** Direct Form I allocates M delays for $x[n]$ and N delays for $y[n]$, totaling $M + N > \max(M, N)$ delays $\implies$ Non-canonical. Direct Form II merges the delay lines via an intermediate state variable $w[n]$, requiring exactly $\max(M, N)$ delays $\implies$ Strictly canonical. (2 Marks)
- **Engineering Selection:** In fixed-point hardware, Direct Form II intermediate node $w[n] = x[n]/A(z)$ suffers from severe resonance peaking and arithmetic overflow when poles are close to the unit circle. Direct Form I attenuates input signals first ($B(z)X(z)$), providing superior dynamic range and lower overflow susceptibility. (2 Marks)